

Supplementary Material

LTSpice Simulation and Dataset Generation Details

S1. LTSpice Circuit Schematics

This section presents the LTSpice configurations used to simulate nanosheet (NS) and forksheet (FS) transistor-based CMOS inverter circuits.

S1.1 Forksheet CMOS Inverter

The forksheet transistor configuration is implemented using a complementary NMOS and PMOS structure. The circuit is driven by a pulsed input voltage and powered by a DC supply.

- NMOS and PMOS devices share parameterized channel length (L) and width (W)
- Drain terminals connected to output node
- Gate terminals driven by pulse source
- PMOS source connected to V_{DD} , NMOS source grounded

S1.2 Nanosheet CMOS Inverter

The nanosheet configuration follows the same inverter topology, with modified device parameters reflecting advanced scaling.

- Reduced channel length and width
- Lower threshold voltage
- Optimized for high-speed operation

S2. Device Parameters and Models

S2.1 Forksheet Parameters

- Channel Length: 60 nm to 180 nm
- Channel Width: 2 μm to 10 μm
- Threshold Voltage: ± 0.7 V
- Transconductance: 150 $\mu\text{A}/\text{V}^2$

S2.2 Nanosheet Parameters

- Channel Length: 2 nm to 10 nm
- Channel Width: 0.25 μm to 2 μm
- Threshold Voltage: ± 0.4 V
- Transconductance: 100 $\mu\text{A}/\text{V}^2$

S2.3 LTSpice Model Definitions

```
.model NMOS_FS NMOS (VT0=0.7 KP=150u)
.model PMOS_FS PMOS (VT0=-0.7 KP=150u)

.model NMOS_NS NMOS (VT0=0.4 KP=100u)
.model PMOS_NS PMOS (VT0=-0.4 KP=100u)
```

S3. Simulation Configuration

S3.1 Voltage Sources

- $V_{DD} = 5$ V (DC supply)
- Input signal:

```
PULSE(0 5 0 0 0 5m 10m)
```

S3.2 Parametric Sweeps

Forksheets:

```
.step param Lval list 180n 150n 120n 90n 60n
.step param Wval list 10u 8u 6u 4u 2u
```

Nanosheet:

```
.step param Lval list 10n 8n 6n 4n 2n
.step param Wval list 2u 1.5u 1u 0.5u 0.25u
```

S3.3 Transient Simulation

```
.tran 100m
```

The simulation captures full switching cycles to analyze transient response and timing behavior.

S4. Waveform Analysis

Each LTSpice simulation generates time-domain output voltage (V_{out}) waveforms.

Key observations:

- Stable switching behavior across all configurations
- Faster transitions for smaller channel dimensions
- Comparable waveform shapes between NS and FS devices

Representative waveform plots should include:

- Small dimension configuration
- Medium configuration
- Large dimension configuration

S5. Data Extraction and Dataset Construction

Raw LTSpice outputs are exported as time-voltage data.

The dataset is constructed by:

1. Extracting $V_{out}(t)$ from simulation output
2. Mapping each waveform to corresponding parameters (L , W , V_{DD} , T , V_{th})
3. Sampling time-domain points into discrete observations
4. Aggregating all configurations into a unified dataset

S6. Physical Modeling Basis

The dataset reflects underlying transistor physics:

$$I \propto \frac{W}{L}(V_{DD} - V_{th})^2 \quad (1)$$

$$\tau \propto \frac{L}{I} \quad (2)$$

These relationships govern switching behavior and are preserved in the dataset.

S7. Variability and Noise Modeling

To improve realism, variability is introduced through:

- Parameter perturbations in L , W , and V_{th}
- Noise in output voltage
- Multiple simulation instances per configuration

This approach captures process variation and enhances model robustness.

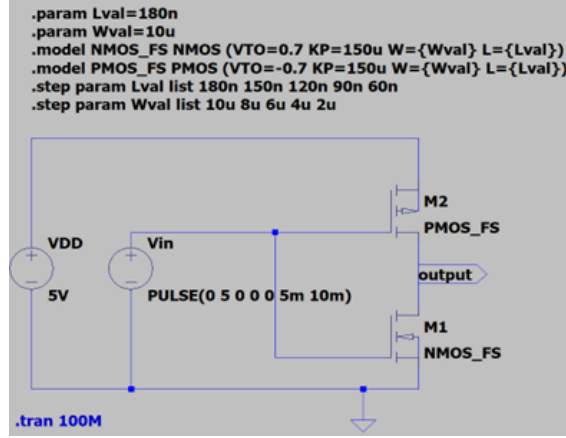


Figure 1: Forksheet (FS) CMOS inverter implemented in LTSpice, showing transistor models, parameter definitions (.model, .param), and parametric sweep configuration (.step param).

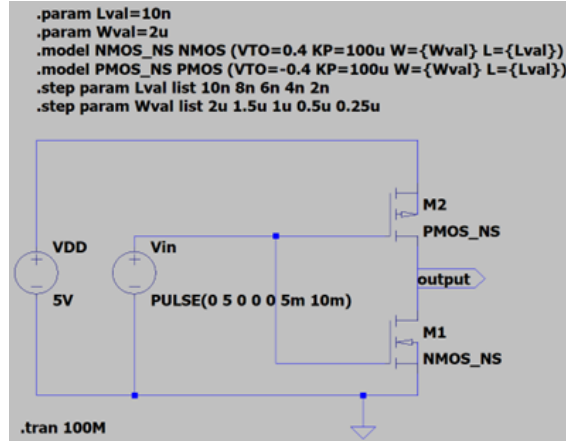


Figure 2: Nanosheet (NS) CMOS inverter implemented in LTSpice, illustrating reduced device dimensions, lower threshold voltage, and corresponding parameter sweep setup.

The circuits are implemented using complementary NMOS and PMOS configurations forming inverter structures driven by a pulsed voltage source. Figures 1 and 2 highlight the parameterized design, including channel dimensions, transistor models, and sweep configurations used to evaluate scaling behavior.

S8. Summary

The LTSpice simulations provide a physics-grounded foundation for dataset generation. The integration of parametric sweeps, transient analysis, and structured data extraction ensures that the dataset accurately represents nanosheet and forksheet transistor behavior across a wide operating range.